Introduction
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Task 1: Automated Backup & Cleanup System

Operating Systems OEL-1

1. Objective

- Task 1: Automated Backup & Cleanup System Objective. Design a script-based solution to manage files/directories with backup and cleanup logic. Your system's /project directory has multiple subdirectories created by different users. Over time it gets cluttered with old .log, .tmp, and large .zip files. Build a shell script that creates a backup_YYYY-MM-DD folder inside /backup. Moves all files older than 30 days from /project into it, preserving the subdirectory structure. Deletes .tmp files older than 7 days directly without backup. Generates a report.txt listing: files moved, files deleted, total space cleared, and any permission errors faced. Logs all actions with timestamps to cleanup.log. Open-ended part: Decide how to handle name conflicts, permissions for other users' files, and what to do if /backup runs out of space. Test with your own sample directory tree. find, mv, cp, rm, date, mkdir -p, du, chmod, >> logging, conditional statements.

2. Environment & Directory Setup

2.1 Sample Directory Tree Creation

```
mkdir -p /home/mehakproject/alice/reports
mkdir -p /home/mehak/project/bob/logs
mkdir -p /home/mehak/project/charlie/temp
mkdir -p /home/mehak/project/shared/archives
mkdir -p /home/mehak/backup
```

```
# Old .log and .zip files (> 30 days old)
touch -t 202502010000 project/shared/archives/backup_old.zip
touch -t 202502150000 project/charlie/temp/cache.zip
touch -t 202502200000 project/alice/reports/q1_summary.log
touch -t 202502250000 project/alice/reports/q2_data.log
touch -t 202503010000 project/bob/logs/server.log

# Old .tmp files (> 7 days old)
touch -t 202604010000 project/alice/reports/draft.tmp
touch -t 202604030000 project/bob/logs/session.tmp
touch -t 202604050000 project/charlie/temp/swap.tmp

# Recent files (< 30 days) — should NOT be touched
touch project/alice/reports/q3_current.log
touch project/bob/logs/today.log
touch project/charlie/temp/recent.tmp
```

2.2 Initial Directory Tree

```
$ find /home/claude/project -type f | sort

/home/mehak/project/alice/reports/draft.tmp
/home/mehak/project/alice/reports/q1_summary.log
/home/mehak/project/alice/reports/q2_data.log
/home/mehak/project/alice/reports/q3_current.log
/home/mehak/project/bob/logs/server.log
/home/mehak/project/bob/logs/session.tmp
/home/mehak/project/bob/logs/today.log
/home/mehak/project/charlie/temp/cache.zip
/home/mehak/project/charlie/temp/recent.tmp
/home/mehak/project/charlie/temp/swap.tmp
/home/mehak/project/shared/archives/backup_old.zip
```

3. Script Design & Implementation

3.1 Variable Declarations & Setup

```
#!/bin/bash

PROJECT_DIR="/home/mehak/project"
BACKUP_BASE="/home/mehak/backup"
LOG_FILE="/home/mehak/cleanup.log"
REPORT_FILE="/home/mehak/report.txt"
BACKUP_MIN_FREE_MB=100

DATE_TAG=$(date +%Y-%m-%d)
BACKUP_DIR="${BACKUP_BASE}/backup_${DATE_TAG}"

FILES_MOVED=0
FILES_DELETED=0
SPACE_CLEARED_KB=0
PERM_ERRORS=0
```

3.2 Logging Function

```
log() {
    echo "[$(date '+%Y-%m-%d %H:%M:%S')] $1" | tee -a "$LOG_FILE"
}
```

3.3 Disk Space Guard (Open-ended: Out-of-Space Handling)

```
FREE_MB=$(df -m "$BACKUP_BASE" | awk 'NR==2 {print $4}')
log "Free space on backup volume: ${FREE_MB} MB"

if [ "$FREE_MB" -lt "$BACKUP_MIN_FREE_MB" ]; then
    log "ERROR: Not enough space. Need ${BACKUP_MIN_FREE_MB} MB, have ${FREE_MB}
MB."
    echo "CRITICAL: Backup aborted — insufficient disk space." >> "$REPORT_FILE"
    exit 2
fi
```

3.4 Conflict Handling (Open-ended: Name Conflicts)

```
# For the backup folder itself:
if [ -d "$BACKUP_DIR" ]; then
    BACKUP_DIR="${BACKUP_DIR}_$(date +%H%M%S)"
    log "Backup folder existed. Using: $BACKUP_DIR"
fi
mkdir -p "$BACKUP_DIR"

# Per-file conflicts inside the backup:
if [ -e "$DEST_PATH" ]; then
    DEST_PATH="${DEST_PATH}.$(date +%s)"
    log "CONFLICT resolved: renaming to $(basename "$DEST_PATH")"
fi
```

3.5 Phase 1 — Delete Old .tmp Files

```
while IFS= read -r -d '' tmpfile; do
    SIZE_KB=$(du -k "$tmpfile" 2>/dev/null | cut -f1)
    if rm "$tmpfile" 2>/dev/null; then
        log "DELETED (tmp): $tmpfile  [${SIZE_KB}K]"
        SPACE_CLEARED_KB=$((SPACE_CLEARED_KB + SIZE_KB))
        FILES_DELETED=$((FILES_DELETED + 1))
    else
        log "PERM ERROR: Cannot delete $tmpfile"
        PERM_ERRORS=$((PERM_ERRORS + 1))
    fi
done < <(find "$PROJECT_DIR" -type f -name "*.tmp" -mtime +7 -print0)
```

3.6 Phase 2 — Move Old Files to Backup

```
while IFS= read -r -d '' oldfile; do
    RELATIVE="${oldfile#$PROJECT_DIR/}"
    DEST_DIR="${BACKUP_DIR}/$(dirname "$RELATIVE")"
    DEST_PATH="${BACKUP_DIR}/${RELATIVE}"

    mkdir -p "$DEST_DIR"

    if [ -e "$DEST_PATH" ]; then
        DEST_PATH="${DEST_PATH}.$(date +%s)"
    fi

    SIZE_KB=$(du -k "$oldfile" 2>/dev/null | cut -f1)

    if mv "$oldfile" "$DEST_PATH" 2>/dev/null; then
        log "MOVED: $oldfile -> $DEST_PATH  [${SIZE_KB}K]"
        SPACE_CLEARED_KB=$((SPACE_CLEARED_KB + SIZE_KB))
        FILES_MOVED=$((FILES_MOVED + 1))
    else
        log "PERM ERROR: Cannot move $oldfile"
        PERM_ERRORS=$((PERM_ERRORS + 1))
    fi
done < <(find "$PROJECT_DIR" -type f ! -name "*.tmp" -mtime +30 -print0)
```

3.7 Phase 3 — Report Generation

```
{
    echo "========================================"
    echo "  Backup & Cleanup Report"
    echo "  Generated : $(date '+%Y-%m-%d %H:%M:%S')"
    echo "  Backup dir: $BACKUP_DIR"
    echo "========================================"
    echo "SUMMARY"
    echo "  Files moved to backup : $FILES_MOVED"
    echo "  Temp files deleted    : $FILES_DELETED"
    printf "  Total space cleared   : %s MB\n" \
        "$(echo "scale=2; $SPACE_CLEARED_KB/1024" | bc)"
    echo "  Permission errors     : $PERM_ERRORS"
    echo "FILES MOVED"
    find "$BACKUP_DIR" -type f | while read -r f; do echo "  $f"; done
    echo "REMAINING IN PROJECT"
    find "$PROJECT_DIR" -type f | while read -r f; do echo "  $f"; done
} > "$REPORT_FILE"
```

4. Script Execution & Terminal Output

```
$ chmod +x cleanup.sh
$ bash cleanup.sh
```

4.1 Terminal Output (cleanup.log contents)

```
[2026-04-16 07:11:43] ====== Cleanup session started ======
[2026-04-16 07:11:43] Free space on backup volume: 10048 MB
[2026-04-16 07:11:44] Backup destination: /home/mehak/backup/backup_2026-04-16
[2026-04-16 07:11:44] --- Phase 1: Deleting .tmp files older than 7 days ---
[2026-04-16 07:11:44] DELETED (tmp): /home/mehak/project/alice/reports/draft.tmp
[0K]
[2026-04-16 07:11:44] DELETED (tmp): /home/mehak/project/charlie/temp/swap.tmp  [0K]
[2026-04-16 07:11:44] DELETED (tmp): /home/mehak/project/bob/logs/session.tmp  [0K]
[2026-04-16 07:11:44] --- Phase 2: Moving files older than 30 days to backup ---
[2026-04-16 07:11:44] MOVED: /home/mehak/project/shared/archives/backup_old.zip
                          ->  /home/mehak/backup/backup_2026-04-
16/shared/archives/backup_old.zip  [500K]
[2026-04-16 07:11:44] MOVED: /home/mehak/project/alice/reports/q2_data.log
                          ->  /home/mehak/backup/backup_2026-04-
16/alice/reports/q2_data.log  [1K]
[2026-04-16 07:11:44] MOVED: /home/mehak/project/alice/reports/q1_summary.log
                          ->  /home/mehak/backup/backup_2026-04-
16/alice/reports/q1_summary.log  [1K]
[2026-04-16 07:11:44] MOVED: /home/mehak/project/charlie/temp/cache.zip
                          ->  /home/mehak/backup/backup_2026-04-
16/charlie/temp/cache.zip  [200K]
[2026-04-16 07:11:44] MOVED: /home/mehak/project/bob/logs/server.log
                          ->  /home/mehak/backup/backup_2026-04-
16/bob/logs/server.log  [1K]
[2026-04-16 07:11:44] --- Phase 3: Generating report ---
[2026-04-16 07:11:44] Report written to /home/mehak/report.txt
[2026-04-16 07:11:44] ====== Cleanup session complete ======
[2026-04-16 07:11:44]   Moved=5 | Deleted=3 | Space=703K | Errors=0
```

5. Generated Outputs

5.1 report.txt

```
========================================
  Backup & Cleanup Report
  Generated : 2026-04-16 07:11:44
  Backup dir: /home/mehak/backup/backup_2026-04-16
========================================

SUMMARY
-------
  Files moved to backup : 5
  Temp files deleted    : 3
  Total space cleared   : .68 MB
  Permission errors     : 0

FILES MOVED
-----------
  /home/mehak/backup/backup_2026-04-16/shared/archives/backup_old.zip
  /home/mehak/backup/backup_2026-04-16/alice/reports/q2_data.log
  /home/mehak/backup/backup_2026-04-16/alice/reports/q1_summary.log
  /home/mehak/backup/backup_2026-04-16/charlie/temp/cache.zip
  /home/mehak/backup/backup_2026-04-16/bob/logs/server.log

REMAINING IN PROJECT (recent files kept)
----------------------------------------
  /home/mehak/project/alice/reports/q3_current.log
  /home/mehak/project/charlie/temp/recent.tmp
  /home/mehak/project/bob/logs/today.log

========================================
```

5.2 Final Directory State

After the script completed, the backup folder preserved the original subdirectory structure:

```
$ find /home/mehak/backup -type f | sort

/home/mehak/backup/backup_2026-04-16/alice/reports/q1_summary.log
/home/mehak/backup/backup_2026-04-16/alice/reports/q2_data.log
/home/mehak/backup/backup_2026-04-16/bob/logs/server.log
/home/mehak/backup/backup_2026-04-16/charlie/temp/cache.zip
/home/mehak/backup/backup_2026-04-16/shared/archives/backup_old.zip
```

Only recent files (less than 30 days old) remained in the project directory:

```
$ find /home/mehak/project -type f | sort

/home/mehak/project/alice/reports/q3_current.log
/home/mehak/project/bob/logs/today.log
/home/mehak/project/charlie/temp/recent.tmp
```